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EDUCATION

- **Harvard University** Boston, MA
M.S. in Health Data Science August 2025 – March 2027
 - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
 - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- **Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 August 2021 – May 2025
 - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, R
- **ML systems:** PyTorch, TensorFlow, scanpy, AnnData, YAML experiment specs
- **Cluster and agents:** SLURM, FASRC Cannon, job arrays, checkpoint/resume, Polar runtime compiler
- **Verification:** unittest, pytest, policy.validate, decoded-frame metrics
- **Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

ENGINEERING PROJECTS

- **Closed-Loop Cluster Autoresearch for scGen vs CellOT** [GitHub] Boston, MA
Sole author — Harvard FASRC Cannon June 2026 – Present
 - Built a self-directed loop (submit → watch → triage → reflect → decide → continue → synthesize) that ran **338** SLURM jobs (318 ok, 20 handled errors) over **79.3** hours on Cannon.
 - Split a deterministic one-delta proposer, gated by `policy.validate()`, from a checkpointed research agenda. Default planner is the rule reasoner. An optional LLM director with token and cost accounting exists and has not been used on recorded runs.
 - Routed tiny-MLP training to CPU partitions because GPU carries 200–500× fairshare cost, and resumed preempted jobs on `serial_requeue` from `cache/last.pt`.
- **Job Search OS. Human-supervised browser automation** [GitHub] Boston, MA
Sole author July 2026 – Present
 - Built a job-search operating system that keeps GitHub as durable policy and evidence, a Google Sheet as Polar runtime state, and Simplify as the application ledger.
 - Compiled Polar workflows from repo YAML so the browser agent reads one runtime file. Polar may Submit only after listed identity and writing gates. Cloud Submit at G2 and G3 stays closed.
 - Kept Computer Use as hands-only on a compiled action sheet, and designed recovery for Mac shutdown, Wi-Fi loss, and browser restart. The system does not send messages or apply on its own.
- **mixhvg-py. Validated Python port of an R HVG ensemble** [GitHub] Boston, MA
Sole author — Scientific software July 2026
 - Ported mixhvg to an AnnData-native API (cells × genes) that combines methods by maximum rank rather than averaging.
 - Validated against R, scran, and Seurat. Seurat vst matches at Spearman **1.0000**. The recommended mixture agrees at Jaccard **0.980** at 500 genes. A weighted median, not a mean, matched R discarded-gene fraction (0.503 vs 0.499).

RESEARCH EXPERIENCE

- **speciesOT. Cross-species single-cell translation** [GitHub] Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute February 2026 – Present
 - Built a hub CLI that catalogs trained runs, ranks them on decoded-space gap-closed rather than raw R^2 , and prints SLURM chains without submitting jobs. Fifteen YAML specs are the intent source of truth.
 - Engineered atlas-scale I/O. Backed AnnData prep so the documented **43 GB** Tabula files are not full-loaded on a login node, and baked gene-axis plus scGen-shift sidecars so a new mouse prediction does not reopen the training atlas.
 - On a paired held-out-species split (LPS / rat-in-train prior, not the Tabula atlas), scGen R^2 was **0.85–0.90** with the source species in training versus **0.65–0.67** when that species was held out. Gene-space ranking reversed with split (6619 orthologs vs 1000 HVGs).